Rational Numbers and Coordinates

Lesson 10 Reflections and Coordinate Signs

Standards 6.NS.C.6b

Learning target I can reflect points across an axis using signs.

Materials Number lines, coordinate grids, rulers, colored pencils.

Lesson Flow 50 Minutes

0–5 Activate prior knowledge. Read the learning target. Ask students what they already know about reflections and coordinate signs. Use a quick model or estimate to make their starting ideas visible.

5–17 Teach and model. Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis. Work through the two examples on the attached instruction page. Have students name the first step before showing it.

17–22 Guided practice. Revisit worked Example 2 with its solution covered. Students propose the next step and explain why it is valid; uncover and compare with the worked solution.

22–45 Independent classwork. Assign the attached 10 questions in order. Check items 1–3 for entry-level understanding, 4–6 for connected procedures, 7–8 for representation and explanation, and 9–10 for application or justification. Ask students to use the instruction-page notes area or extra grids for longer work.

45–50 Check understanding. Use classwork question 10 as an exit discussion: Plot (-8, -7), reflect it across both axes, and give the reflected coordinates. Compare the result with the answer key and ask a student to defend the method.

Support and Extension

Support Model item 1 together with the listed tools, then have students attempt item 2 independently. Watch order versus absolute value, coordinate order, and which sign changes in a reflection.

Extension Ask students to solve a final item another way, draw a second representation, or create a related problem with a checked answer.

Next step Record which students can explain the target independently. Begin the next teaching session with a brief correction for any recurring error.

Name _____ Date _____

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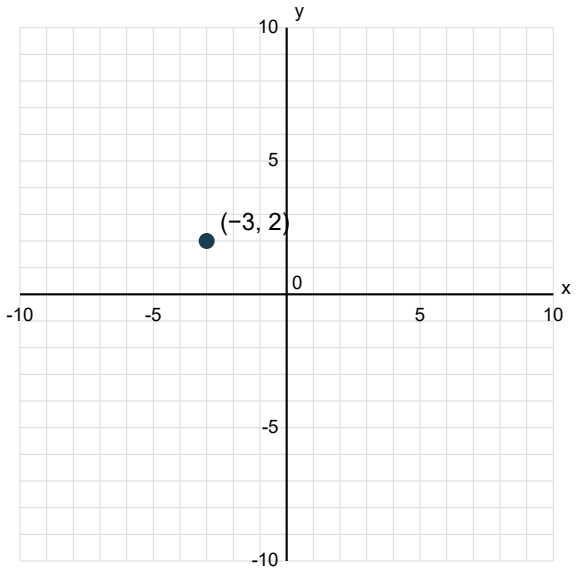
Learn and Discuss

Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis.

Example 1 Plot (-7, -3), reflect it across y-axis, and give the reflected coordinates.
Solution (7, -3). Reverse the x sign: (7, -3).
Example 2 Plot (-4, -7), reflect it across both axes, and give the reflected coordinates.
Solution (4, 7). Reverse both signs: (4, 7).

Classwork Show Your Work

Complete the 10 problems on the following page. Refer to the examples, draw models, and check your results.



Model a point (−3, 2) means move 3 left and 2 up from the origin. Label the axes and scale before plotting.

Lesson 10 Reflections and Coordinate Signs

Name _____ Show calculations, models, and units.

1. Plot (-8, 6), reflect it across x-axis, and give the reflected coordinates.

2. Plot (-2, -5), reflect it across y-axis, and give the reflected coordinates.

3. Plot (-5, 6), reflect it across x-axis, and give the reflected coordinates.

4. Plot (-4, -2), reflect it across y-axis, and give the reflected coordinates.

5. Plot (-7, 4), reflect it across x-axis, and give the reflected coordinates.

6. Plot (-2, -7), reflect it across y-axis, and give the reflected coordinates.

7. Plot (-2, 3), reflect it across both axes, and give the reflected coordinates.

8. Plot (-7, -3), reflect it across both axes, and give the reflected coordinates.

9. Plot (-6, 2), reflect it across both axes, and give the reflected coordinates.

10. Plot (-8, -7), reflect it across both axes, and give the reflected coordinates.

Reflections and Coordinate Signs

Rational Numbers and Coordinates

1. (-8, -6)
Reasoning Reverse the y sign: (-8, -6).

2. (2, -5)
Reasoning Reverse the x sign: (2, -5).

3. (-5, -6)
Reasoning Reverse the y sign: (-5, -6).

4. (4, -2)
Reasoning Reverse the x sign: (4, -2).

5. (-7, -4)
Reasoning Reverse the y sign: (-7, -4).

6. (2, -7)
Reasoning Reverse the x sign: (2, -7).

7. (2, -3)
Reasoning Reverse both signs: (2, -3).

8. (7, 3)
Reasoning Reverse both signs: (7, 3).

9. (6, -2)
Reasoning Reverse both signs: (6, -2).

10. (8, 7)
Reasoning Reverse both signs: (8, 7).